

118. ANTIOXIDANTS FOR ENDURANCE PERFORMANCE: WHEN THEY HELP AND WHEN THEY GET IN THE WAY

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Endurance training is, at its core, a stress-and-adaptation process. This stress is ***supposed*** to happen. It signals your cells to get stronger by building more mitochondria, improving muscle efficiency, and positively refining how your body uses fuel during exercise. **Every hard ride creates small amounts of muscle damage, metabolic stress, and the production of reactive oxygen species (ROS).** Reactive oxygen species are byproducts of normal metabolism. They are highly reactive molecules formed when oxygen is used to produce energy. During exercise, oxygen demand is high and more ROS are generated inside the working muscles.

While large amounts can damage cells, moderate increases in are actually useful because they act as signals that help the body adapt by upregulating its own protective systems and stimulating the production of new mitochondria. This is where antioxidants come in. Antioxidants are molecules that can neutralize, or “scavenge,” excess ROS before they cause damage. They include both dietary compounds like vitamins C and E, carotenoids, and polyphenols from fruits and vegetables, as well as antioxidant enzymes that the body produces itself. **The balance between ROS and antioxidants determines whether oxidative stress leads to harm or to beneficial adaptation.**

Antioxidant supplements often come up in the context of training because they can reduce oxidative stress. But not all antioxidant strategies are equal. And in some cases, too much antioxidant intake (especially from supplements) can actually *blunt* the signals we’re trying to enhance through training.

Let’s break down how this works, and how cyclists can use antioxidants strategically without getting in their own way.

1. What Actually Happens During Hard Training

During exercise, your muscles produce ROS (“oxidants”). These are often portrayed as “bad,” but that’s quite misleading. ROS serve as one of the *messengers* that turn on training-related adaptations—especially mitochondrial biogenesis and improvements in endurance capacity. This is what we want!

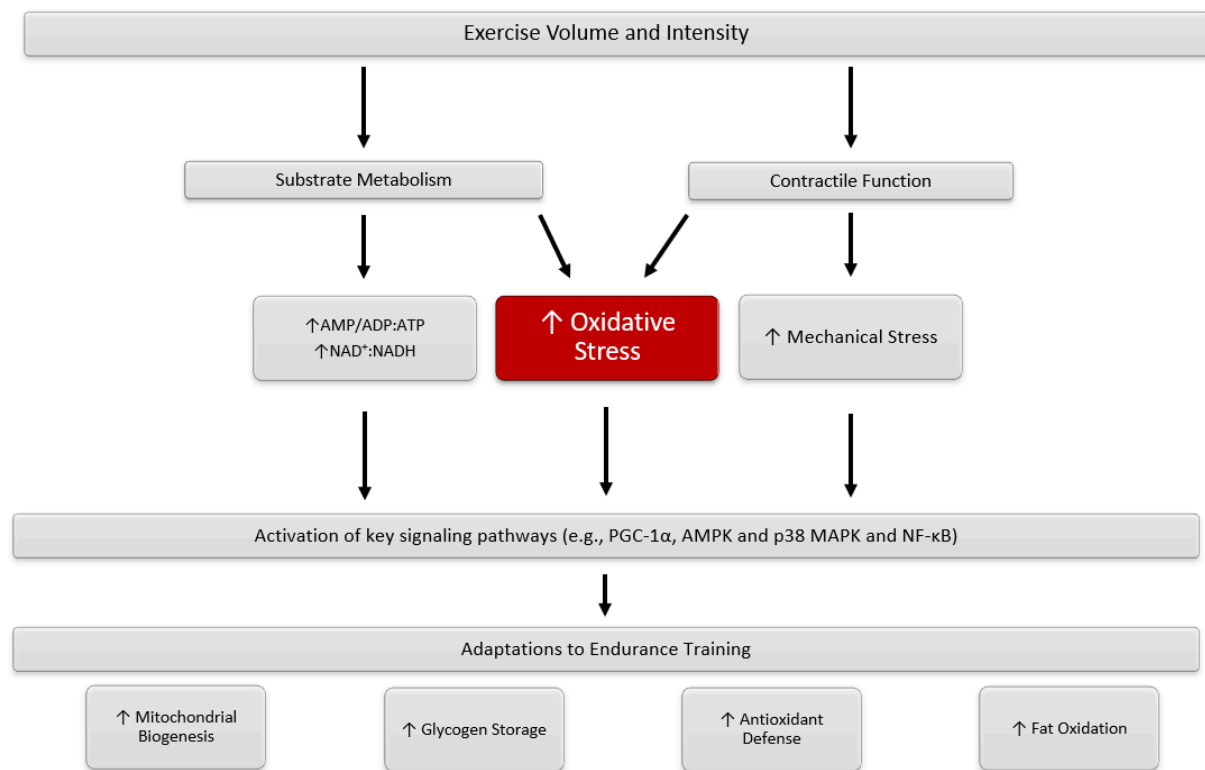


Figure 1. Big-picture overview showing how an exercise stimulus (volume and intensity) impacts adaptation signals through multiple different pathways including substrate metabolism (i.e. what we’re burning) and muscle contraction (repeated over and over). Oxidative stress is one of several key signals.

This process involves several intracellular signalling pathways, including:

- **PGC-1α** (drives mitochondrial growth)
- **AMPK and p38 MAPK** (respond to metabolic stress)
- **NF-κB** (involved in inflammatory and antioxidant defense gene expression)

When ROS rise during exercise, they activate these pathways, helping your body “learn” how to handle stress more efficiently over time.

Chronic exercise itself is associated with a significant increase in antioxidant capacity and reduction in pro-oxidant parameters, regardless of intensity, volume, type of exercise and studied population [1]. In fact, a study elite rowers found that the number of years of training had a larger effect on antioxidant status than dietary intake [2]!

However—too much oxidative stress *can* contribute to excessive fatigue or slow recovery during dense training blocks. So, the goal is not “remove all ROS,” but *maintain balance*.

2. How Antioxidants Work

1. Scavenging Free Radicals (Direct Neutralization)

This is what most people think of: antioxidants bind and neutralize free radicals before they can damage cells. Supplements like Vitamin C, Vitamin E, and some polyphenol extracts work mainly in this direct scavenging role.

2. Upregulating Endogenous Defense Systems (Building Your Own Shield)

Your cells also have internal antioxidant systems—superoxide dismutase, catalase, glutathione, etc. These can be *upregulated* by certain nutritional and training stimuli. This process is mediated through transcription factors, especially NF-κB, which increases the expression of your cells' own defense enzymes. This internal defense system can be stimulated by plant foods rich in polyphenols (berries, cocoa, olive oil, coffee, tea), and even by exercise itself, not by eliminating ROS, but by helping your body *handle* them better long-term.

This distinction is critical:

Scavenging eliminates the signal. Upregulating strengthens your response.

And for endurance athletes, we want the *response*, not the removal of the signal. Several studies have shown that adaptations to endurance training largely depend on exercise-induced oxidative stress [3, 4].

3. Where Supplementation Can Go Wrong

Some antioxidant supplements—particularly high-dose vitamin C or E—have been shown to blunt endurance training adaptations. They reduce the ROS signal so much that pathways like PGC-1α and AMPK are activated to a lesser degree, which may reduce mitochondrial adaptations over time.

This effect has been noted in research examining supplement use during endurance training programs, where vitamin C and E supplementation did not improve performance and in some cases attenuated adaptations such as improvements in mitochondrial enzymes and VO₂max [5].

Similarly, antioxidant supplementation during short-term recovery periods is discussed in post-exercise recovery research, where it may reduce muscle damage markers but at the potential cost of limiting adaptive remodeling in the long term [6].

4. So Should Cyclists Avoid Antioxidants? Not Exactly.

The key is *timing* and *source*.

Best case for antioxidants:

- Heavy training blocks with accumulated fatigue
- Multi-day stage races
- Back-to-back hard sessions where recovery matters more than adaptation
- Illness, high travel stress, or poor sleep

Worst case for antioxidants:

- Base phase
- When targeting aerobic/mitochondrial development
- Immediately post-workout when adaptation signals are highest

If you blunt the stress signal *when you want to get fitter*, you may adapt less.

If you blunt the stress signal *when you just want to survive tomorrow's race*, that's strategic.

5. Food First, Then Supplements

Research suggests whole-food antioxidants *do not* interfere with training adaptations the same way isolated supplements may. This is likely because whole foods contain compounds that *support* ROS signaling regulation rather than simply blocking it.

Great choices include:

- Berries
- Citrus fruit
- Leafy greens
- Olive oil
- Tomatoes
- Nuts
- Dark chocolate
- Coffee and tea

Think of these as building your *internal antioxidant capacity*—not removing the training signal.

Supplements (vitamin C, vitamin E, NAC, resveratrol, curcumin, etc.) should be used purposefully, not automatically.

Situation	Best Strategy	Avoid
Base Phase / Aerobic Build	Let ROS signaling occur → focus on whole foods	Daily antioxidant supplements
High-Intensity Training Block	Food antioxidants + good recovery fueling	High-dose vitamin C/E
Stage Races / Double Days	Strategic supplements <i>after</i> hard sessions to speed recovery	
Travel / Illness Risk	Supplements may help immunity	

Supplement	Typical Dose Studied	Duration / Context	Key Findings	Practical Takeaway
Vitamin C	500–1000 mg · day ⁻¹	6–8 weeks during endurance training	High doses may blunt mitochondrial and VO ₂ max adaptations despite reduced oxidative stress	Avoid chronic use during base/build; short-term use may help recovery during stage races or illness.
Vitamin E (α-tocopherol)	400–800 IU · day ⁻¹	6–8 weeks	Combined C + E supplementation can reduce ROS signaling (PGC-1α, AMPK) and cellular adaptation	Similar caution as for vitamin C—occasional use for acute recovery only.
Polyphenols (e.g., quercetin, resveratrol, catechins)	500–1000 mg polyphenols · day ⁻¹	≥3 days around exercise or 4–8 weeks training	Can upregulate NRF2, SIRT1–AMPK–PGC-1α pathways and increase fat oxidation, though some (e.g., resveratrol) may blunt VO ₂ max gains in sedentary subjects	Favor whole-food sources (berries, tea, cocoa); supplemental use may assist recovery but not daily adaptation.
Curcumin / Tart Cherry / Pomegranate	~500–1000 mg extract · day ⁻¹ (≈ 1000 mg polyphenols)	3–7 days pre-competition or post-exercise	Shown to reduce muscle soreness and inflammation without harming adaptation	Effective recovery aid during multi-day events or high stress.

Thanks for Reading!

Share with your Team!

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You can find Jeff and many interesting things he is working on his website [here](#).

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